Damping Accumulator – Maintenance Practices

1. Removal/Installation

   A. Remove Airframe System Accumulator

      1) Discharge airframe system accumulator hydraulic pressure by operating wing flaps.

      2) Open left and right-hand access panels on underside of nose section. (Refer to 12-00-00, Servicing – General Data).

      3) Discharge all air pressure from airframe system accumulator by loosening air valve approximately one-quarter turn and waiting until all air has escaped. Check indicator on accumulator reads zero.

         WARNING FAILURE TO PROPERLY DISCHARGE ALL AIR PRESSURE FROM AN ACCUMULATOR MAY RESULT IN INJURY TO PERSONNEL AND/OR DAMAGE TO EQUIPMENT.

      4) Disconnect air line at head of accumulator. Cap and plug open connection and line.

      5) Place container to receive spillage. Remove nuts and release two clamps, and unscrew accumulator with union from crosspiece.

   B. Install Airframe System Accumulator

      1) If new accumulator is to be installed, transfer fitting from unserviceable accumulator and use new O-rings.

      2) Install new O-ring at crosspiece, and screw accumulator union into crosspiece.

      3) Attach accumulator to tray with two clamps and nuts.

         CAUTION OVERTIGHTENING OF ACCUMULATOR MOUNTING CLAMPS COULD RESULT IN DAMAGE TO ACCUMULATOR.

      4) Overcome run down torque of clamp nuts then add 15 – 20 pound–inches torque maximum to seat nut.

      5) Connect air line at head of accumulator.

      6) Charge accumulator with dry nitrogen to 750 ± 50 psi.

      7) Check reservoir fluid level. Function test hydraulic power package. (Refer to 29-10-00, Hydraulic Power Package (Mod 6/2039) – General Data).

      8) Bleed hydraulic handpump. (Refer to 29-20-11, Hydraulic Handpump (Mod 6/2093) – Maintenance Practices).

      9) Bleed nose wheel steering and wing flaps systems. (Refer to 32-50-00, Nose Wheel Steering System – General Data and 27-50-00, Wing Flaps System – General Data).

      10) Close left and right-hand access panels. (Refer to 12-00-00, Servicing – General Data).

2. Adjustment/Test

   Equipment required:

   - Manometer or U tube
   - Suitable hydraulic test rig
   - Suitable water container

A. Bench Test (Non Destructive) Airframe System Accumulator

1) Remove accumulator from hydraulic power package (refer to Para 1.A).

2) Remove fittings from hydraulic and gas end caps.

3) Inspect accumulator for obvious damage.

4) Do a fluorescent penetrant inspection on hydraulic and gas end caps. Check for imperfections.

5) Do a magnetic particle inspection on shell. Check for imperfections.

6) Operate piston two times in each direction using hydraulic fluid, nitrogen gas or dry air, to a maximum pressure of 50 psi.

7) Check that piston moves through entire length of shell without seizing or binding.

Note Do not operate piston manually.

8) With the piston in the balanced position (at least one inch from each end cap), fill hydraulic end and air end with fluid and plug one end. Attach a suitable pressure indicator to other end.

9) Ensure all air is removed from accumulator and lines.

10) Apply 2250 psi hydraulic pressure to the accumulator and hold for three minutes. Check that no external leakage or damage exists.

11) Release pressure and remove fluid from accumulator.

12) Connect a nitrogen or dry air source to gas end cap port. Slowly apply pressure until piston bottoms lightly on hydraulic end cap.

13) Connect port of hydraulic end cap to manometer or one limb of U tube.

14) Apply gas pressure of 2000 ± 100 psi to gas end cap port, and hold for minimum of three minutes. Internal air leakage must not exceed 0.10 milliliters.

15) Reduce pressure to 200 ± 25 psi, and hold for three minutes. Internal air leakage must not exceed 0.10 milliliters. Release pressure.

16) Connect nitrogen or dry air source to both gas and hydraulic end cap ports.

17) Prepare to submerge accumulator in water.

CAUTION **EXERCISE CARE TO PREVENT WATER FROM ENTERING AND MAKING CONTACT WITH INTERNAL SURFACES OF ACCUMULATOR BY APPLYING GAS PRESSURE PRIOR TO SUBMERSION, AND MAINTAINING UNTIL AFTER WITHDRAWAL FROM WATER.**

18) Apply gas pressure of 3000 ± 100 psi to hydraulic and gas end cap ports simultaneously and submerge accumulator in water. Hold pressure for five minutes.

19) Check that no external leakage exists.

20) Reduce pressure to 200 ± 25 psi and hold for five minutes.

21) Check that no external leakage exists.

22) Remove accumulator from water and slowly release pressure.

23) If required for service, dry, fill with hydraulic fluid (MIL–H–5606) and reinstall in hydraulic power package. If not required for immediate use, fill with hydraulic fluid (MIL–H–6083 Type 1), and cap ports.

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